

6.1: Reciprocal Trig Functions:

$(\csc\theta =)\ \operatorname{cosec}\theta = \frac{1}{\sin\theta}$

$\sec\theta = \frac{1}{\cos\theta}$

$\cot\theta = \frac{1}{\tan\theta}$

$\sin^2\theta + \cos^2\theta = 1$

$\Downarrow$ $\qquad$ $\Downarrow$

$\frac{\sin^2\theta + \cos^2\theta}{\cos^2\theta} = \sec^2\theta = \tan^2\theta + 1$ $\qquad$ $\frac{\sin^2\theta + \cos^2\theta}{\sin^2\theta} = \operatorname{cosec}^2\theta = 1 + \cot^2\theta$

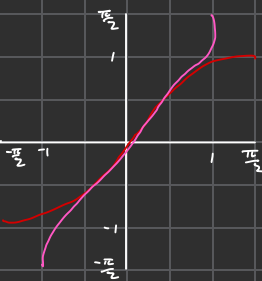
6.2: Working with Trig. Equations & Identities

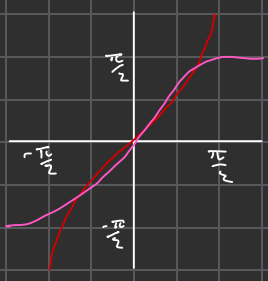
For inverse functions, a restricted domain & range is required, for the function to be a valid mapping

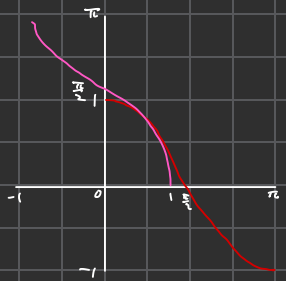
$y = \sin\theta \Rightarrow -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ $\qquad$ $-\frac{\pi}{2} \leq \arcsin x \leq \frac{\pi}{2}$

$y = \cos\theta \Rightarrow 0 \leq \theta \leq \pi$ $\qquad$ $0 \leq \arccos x \leq \pi$

$y = \tan\theta \Rightarrow -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$ $\qquad$ $-\frac{\pi}{2} \leq \arctan x \leq \frac{\pi}{2}$







$y = \sin x$ $\quad y = \arcsin x$ $\qquad$ $y = \tan x$ $\quad y = \arctan x$ $\qquad$ $y = \cos x$ $\quad y = \arccos x$

6.3: Equations with Radians

For trigonometric equations, use substitution to find a solution, adjusting the bounds appropriately, then find in terms of the original variable

Find all possible valid solutions relating to a unit circle, by using symmetry or periods (for $\tan\theta$)